DIAGNOSING JEPA WORLD MODELS WITH ACTION-CONDITIONED PREDICTIVE CONSISTENCY

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ABSTRACT

Joint-embedding predictive architectures (JEPAs) learn world models that predict in a compact latent space rather than in pixels, reducing the pressure to model nuisance appearance. Yet this provides no guarantee against visual perturbations: they can still alter the encoded representation and affect subsequent action-conditioned predictions. Bisimulation captures this requirement precisely: two observations should be treated as the same state only when their action-conditioned consequences agree. Guided by this criterion, we introduce Action-Conditioned Predictive Consistency (ACPC), a diagnostic that measures how far a clean history and a visually perturbed view of it diverge after being rolled forward under the same action sequence. We prove that this divergence bounds the perturbation-induced change in multi-step prediction error and planner cost. Building on pairwise ACPC, we define two complementary measures: the Invariance Radius (IR) summarizes clean–perturbed rollout spread, while the Separation Rate (SR) checks whether different states remain distinguishable after rollout. Experiments on four visual control tasks show that pairwise ACPC predicts perturbation-induced prediction and cost changes. On LeWM, the IR–SR screen transfers across tasks, and both IR and SR remain informative under blur and resize. PLDM exhibits similar diagnostic trends under a different architecture.

1 INTRODUCTION

World models support control by predicting the consequences of candidate actions (Hafner et al., 2025; Hansen et al., 2024). Joint-embedding predictive architectures (JEPAs) do so in latent space, avoiding the need to reconstruct nuisance appearance (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024). Such representations should be insensitive to task-irrelevant visual perturbations while preserving distinctions between situations that require different actions. This requirement echoes the intuition behind bisimulation, which defines state equivalence through action-conditioned consequences rather than visual appearance (Gelada et al., 2019; Zhang et al., 2021). Yet a small input change can still alter the encoded history and then grow or contract through the learned dynamics. Encoder distance therefore misses the quantity that matters to a latent planner—the separation between the resulting predicted trajectories.

We introduce *Action-Conditioned Predictive Consistency* (ACPC), the distance between clean and perturbed latent rollouts generated under the same action sequence. The *Invariance Radius* (IR) summarizes this sensitivity across histories. Low IR alone admits a constant representation, so the *Separation Rate* (SR) checks whether selected different-state rollouts remain outside the perturbation radius. Figure 1 shows the construction.

ACPC gives samplewise bounds on perturbation-induced changes in multi-step prediction error and fixed-candidate planning cost. On four control tasks, recorded-action ACPC predicts error drift better than short-horizon and action-destroyed controls; planner-horizon ACPC adds information about cross-entropy method (CEM) selection regret (Kroese et al., 2006) beyond the one-step planning baseline. At checkpoint level, IR follows recovery in the original Gaussian sweep. SR is nonbinding there, but becomes an active guard as evaluation severity increases under the same frozen rule. We also test the diagnostics under blur, resize, and a second architecture.

Contributions. We (i) define a target-free, action-conditioned pairwise diagnostic and offline IR/SR checkpoint summaries; (ii) derive error, cost, and fixed-pool selection bounds; and (iii) connect these

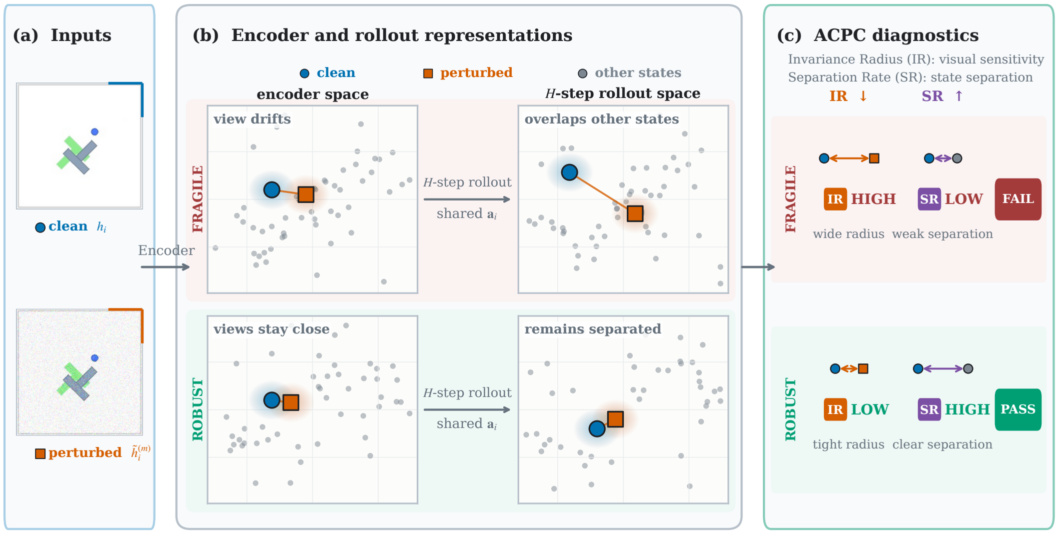

Figure 1: **ACPC compares predicted consequences, not pixels. (a–b)** A frozen model rolls a clean history and its perturbed view forward under the same actions; ACPC is the distance between the two latent trajectories. **(c)** IR summarizes clean–perturbed spread, whereas SR checks whether tested different-state trajectories remain beyond that spread. The geometry is schematic.

quantities to prediction and planning behavior across tasks, perturbations, training strengths, and model families. The checkpoint screen is an offline diagnostic for the evaluated shift and state-pair catalog, not a closed-loop guarantee.

2 Related Work

World Models. DreamerV3 and TD-MPC2 use learned dynamics for imagined control and latent planning (Hafner et al., 2025; Hansen et al., 2024). JEPAs instead learn by predicting representations rather than reconstructing pixels (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024). LeWM, our primary model family, adds action-conditioned latent prediction; Planning with a Latent Dynamics Model (PLDM) provides a second architecture (Maes et al., 2026; Sobal et al., 2025). Prior work analyzes collapse, action conditioning, and which factors predictive objectives retain (Tang et al., 2023; Khetarpal et al., 2025; Littwin et al., 2024). We study the complementary post-training question: how a visual perturbation propagates through a frozen predictor.

Robustness to Visual Perturbations. Data augmentation improves pixel-based control (Yarats et al., 2021; 2022), while auxiliary representation losses target visual generalization (Hansen & Wang, 2021). World-model methods also isolate task-relevant or controllable factors (Fu et al., 2021; Wang et al., 2022; Pan et al., 2022), including through temporal masking with bisimulation (Sun et al., 2024). Bisimulation formalizes behavioral similarity through rewards and action-conditioned transitions (Gelada et al., 2019; Zhang et al., 2021; Shimizu & Tomizuka, 2025), including a reward-free formulation based only on transitions (Toso et al., 2026). MWM enforces action-conditioned rollout consistency during post-training (Yan et al., 2026). ACPC uses the same consequence-based perspective as a post-hoc diagnostic, without retraining the model.

Diagnostics for World Models. One-step model accuracy can be a poor proxy for control (Lambert et al., 2020). Recent work therefore probes action content, rollout validity, or downstream decisions (Chen, 2026; Zhang et al., 2026b; Ruan et al., 2026; Yeom et al., 2026). Attack benchmarks instead search for closed-loop failures (Guo et al., 2026; Zhang et al., 2026a). Other work compares predicted multi-step latent transitions with those produced by the environment (Vakalis, 2026), or compares predicted and true plan costs (You et al., 2026). Certification methods bound policy actions or return under observation uncertainty (Lütjens et al., 2020; Wu et al., 2022); our narrower bounds concern the winner or elite set for one evaluated pair and a fixed candidate pool. ACPC fills a distinct gap: it measures perturbation propagation through a shared- action latent rollout without requiring the true future.

3 Action-Conditioned Predictive Consistency as a Diagnostic

We first define pairwise ACPC and its error and planning-cost consequences, then aggregate it into the checkpoint-level IR and SR diagnostics.

3.1 Pairwise ACPC

Let h and $\tilde{h}$ be a clean history and its visually perturbed view, E_θ a frozen encoder, and F_θ a frozen action-conditioned predictor. Under the shared action sequence $\mathbf{a} = (a_0, \ldots, a_{H-1})$, their step-$k$ predictions are

$$\hat{z}_k = F_\theta^k(z, \mathbf{a}_{0:k-1}), \qquad \hat{\tilde{z}}_k = F_\theta^k(\tilde{z}, \mathbf{a}_{0:k-1}). \tag{1}$$

Let Π map predictions to the latent space in which planner costs are evaluated, and let nonnegative weights satisfy $\sum_k \alpha_k = 1$. We define

$$\mathrm{ACPC}_H(h, \tilde{h}, \mathbf{a}) = \left(\sum_{k=1}^{H} \alpha_k \left\| \Pi(\hat{z}_k) - \Pi(\hat{\tilde{z}}_k) \right\|_2^2 \right)^{1/2}, \tag{2}$$

and denote the corresponding weighted rollout vector by $\bar{G}_{\mathbf{a}}(E_\theta(h))$. Unlike encoder shift, ACPC measures the difference after prediction; it measures consistency, not accuracy. We use uniform weights and $H = 8$ for logged trajectories, and $H = 5$ for the CEM analysis.

3.2 Prediction-Error Bounds

ACPC itself needs no observed future. For analysis, let $Y_{\mathbf{a}}^H$ be the logged future in the same weighted space. Both rollouts share this target.

Proposition 1 (Prediction-error change bound)**.** *For $e_h = \|\bar{G}_{\mathbf{a}}(E_\theta(h)) - Y_{\mathbf{a}}^H\|_2$ and the analogous $e_{\tilde{h}}$, every paired sample satisfies*

$$|e_{\tilde{h}} - e_h| \leq \mathrm{ACPC}_H(h, \tilde{h}, \mathbf{a}). \tag{3}$$

The reverse triangle inequality proves the result (Appendix A). It bounds error *change*, including $(e_{\tilde{h}} - e_h)_+$, not absolute prediction error.

3.3 Selection Stability from Planning-Cost Bounds

For candidate action sequence $\mathbf{a}^j$, let

$$x_j = \Pi(F_\theta^H(E_\theta(h), \mathbf{a}^j)), \qquad \tilde{x}_j = \Pi(F_\theta^H(E_\theta(\tilde{h}), \mathbf{a}^j))$$

be the clean and perturbed predicted endpoints. Their displacement $r_j = \|x_j - \tilde{x}_j\|_2$ satisfies

$$r_j \leq \frac{\mathrm{ACPC}_H(h, \tilde{h}, \mathbf{a}^j)}{\sqrt{\alpha_H}}.$$

Proposition 2 (Planning-cost bounds)**.** *Suppose the planner scores candidate j by its squared distance to a fixed goal embedding g:*

$$C_j = \|x_j - g\|_2^2, \qquad \tilde{C}_j = \|\tilde{x}_j - g\|_2^2.$$

Define the candidate-specific cost-change bound

$$b_j = r_j \big(\|x_j - g\|_2 + \|\tilde{x}_j - g\|_2 \big). \tag{4}$$

Then $|\tilde{C}_j - C_j| \leq b_j$.

Let w and $\tilde{w}$ be the winners under the clean and perturbed inputs. If the winner changes, the perturbed winner's excess cost under the clean prediction satisfies

$$0 \leq C_{\tilde{w}} - C_w \leq b_{\tilde{w}} + b_w. \tag{5}$$

The endpoint factor in b_j is observable and r_j is bounded by candidate-specific ACPC. If the clean winner's gap to every competitor exceeds $b_w + b_j$, the winner is preserved; analogous inequalities preserve a top-k elite set (Proposition 3). For paired CEM runs with common random numbers, preserving the elite set at every round keeps all later pools and the returned action aligned (Corollary 1). Once an elite set differs, this fixed-pool guarantee no longer applies. It concerns the model-based choice, not environment return.

3.4 Checkpoint-Level IR and SR

For n logged anchors, we perturb history h_i independently M times and compute ACPC under its recorded actions. To remove history-dependent scale, we divide by s_i, the median one-step clean motion over the next H observed transitions (Appendix C):

$$R_i^{(m)} = \frac{\mathrm{ACPC}_H(h_i, \tilde{h}_i^{(m)}, \mathbf{a}_i)}{s_i + \varepsilon_{\mathrm{norm}}}, \qquad \bar{R}_i = \frac{1}{M}\sum_{m=1}^{M} R_i^{(m)}. \tag{6}$$

The raw IR is the upper quantile of the draw-averaged sensitivities,

$$\mathrm{IR}_q^{\mathrm{raw}}(\theta) = Q_q\big(\{\bar{R}_i\}_{i=1}^n\big). \tag{7}$$

with $q = 0.90$ in our protocol. Lower IR means that perturbed views remain closer to their clean rollouts.

For SR, each eligible anchor is paired with a nearby history having a different state-coordinate label. Both are rolled out under the anchor's actions, and their normalized distance is D_i^{diff}. SR is the fraction that exceeds raw IR by margin δ:

$$\mathrm{SR}_{q,\delta}(\theta) = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} \mathbf{1}\big[D_i^{\mathrm{diff}} > \mathrm{IR}_q^{\mathrm{raw}}(\theta) + \delta\big], \tag{8}$$

We set $\delta = 0.10$. Constant representations have zero different-state distance and fail SR, while high SR alone does not imply low perturbation sensitivity. SR therefore complements rather than replaces IR, and applies only to the tested label catalog. Appendix C gives the full normalization and pairing protocol.

3.5 Checkpoint Screening

SR uses raw IR because it compares distances within the same checkpoint. To compare a trained checkpoint with its unaugmented reference, we use relative IR. Let θ_0 denote the unaugmented checkpoint from the same task, training run, and model family. We define

$$\mathrm{IR}_q^{\mathrm{rel}}(\theta;\theta_0) = \frac{\mathrm{IR}_q^{\mathrm{raw}}(\theta)}{\mathrm{IR}_q^{\mathrm{raw}}(\theta_0)}. \tag{9}$$

Relative IR equals 1 for the reference checkpoint. Values below 1 indicate lower sensitivity than the reference, while values above 1 indicate higher sensitivity. This comparison is defined within the same task, training run, and model family; it does not make IR directly comparable across them.

A checkpoint passes only if $\mathrm{IR}_q^{\mathrm{rel}} \le t_{\mathrm{IR}}$ and $\mathrm{SR}_{q,\delta} \ge t_{\mathrm{SR}}$. We choose both thresholds on source tasks and apply them unchanged to held-out tasks, separately for each model family. Planning success selects and evaluates the rule but is not an input to ACPC, IR, or SR. The resulting checkpoint-level findings are empirical; they do not follow from the pairwise bounds.

4 Experiments

We test ACPC against pair-level prediction and planning consequences, then evaluate IR and SR across checkpoints, tasks, visual shifts, and model families.

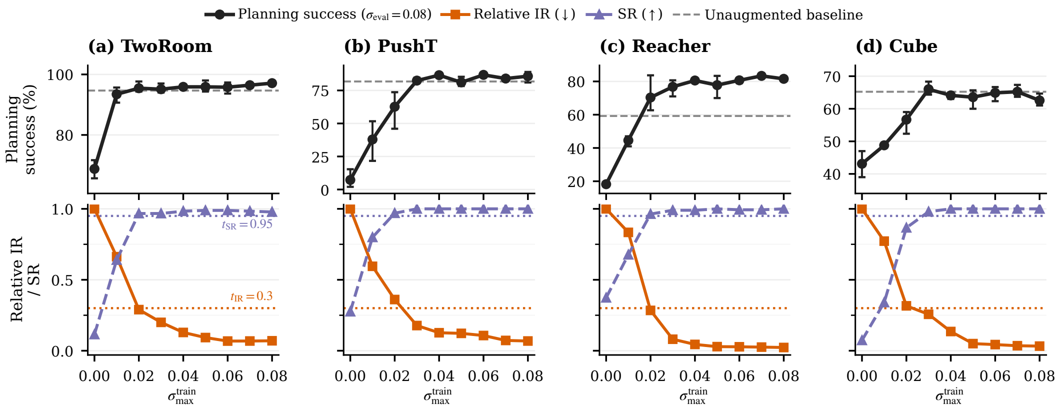

Figure 2: Planning success and diagnostics across the original Gaussian training sweep at evaluation noise $\sigma = 0.08$. Points average three training runs; success bars span their range. Dashed lines show unaugmented clean success, and dotted lines show the fixed IR/SR thresholds. Lower IR and higher SR are favorable.

4.1 Evaluation Protocol

Models, Tasks, and Evaluation. Our primary model is LeWM; PLDM provides a second architecture (Maes et al., 2026; Sobal et al., 2025). We evaluate TwoRoom navigation, PushT manipulation, Reacher control, and OGBench-Cube manipulation. CEM plans five steps ahead, and success is the percentage of episodes reaching the goal. The main sweep evaluates Gaussian noise at $\sigma = 0.08$ and trains one unaugmented plus eight augmented checkpoints ($\sigma_{\text{max}} \in \{0.01, \ldots, 0.08\}$). Each task–condition pair has three training runs; each checkpoint has three 100-episode evaluation seeds. The training run is the replication unit. We also test blur ($k = 15$) and resize (scale 0.25), always perturbing the history while keeping the goal clean.

Diagnostic Protocol. The frozen diagnostic uses 100 anchors, five perturbation draws, eight-step uniformly weighted rollouts, draw averaging, and a q90 anchor summary. SR uses nearby histories with different endpoint-state labels and margin 0.10. Appendix C specifies pairing, normalization, and thresholds.

Broad-severity protocol. In a separate LeWM experiment, we cross $\sigma_{\text{max}} \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.7, 1.0, 1.5\}$ with $\sigma_{\text{eval}} \in \{0.08, 0.16, 0.32, 0.64\}$ over three runs. Training samples each frame's noise from $\text{Uniform}(0, \sigma_{\text{max}})$. The rollout, pair catalog, normalization, and thresholds are frozen. We analyze this 8×4 factorial separately from the original sweep.

Threshold Protocol. A checkpoint recovers when it reaches 80% of the improvement observed across the fixed nine-checkpoint sweep while losing at most five clean-success points; the levels meeting this criterion define a recovery range. We select IR/SR thresholds on every nonempty proper subset of tasks and apply them to the remaining tasks (14 directional partitions), averaging runs within task and weighting tasks equally. Blur and resize do not enter threshold selection; their component analysis uses 24 matched checkpoint pairs without shift-specific calibration.

4.2 IR and SR across Checkpoint Recovery

At $\sigma_{\text{eval}} = 0.08$, unaugmented LeWM loses 22.1–74.6 success points across tasks. Augmentation recovers performance over task-dependent ranges (Figure 2). Reacher also improves clean success from 59.2% to 76–82%, so its noisy recovery is not attributable to robustness alone.

Every checkpoint meeting the recovery criterion has lower IR and higher SR than its unaugmented reference. Across 108 checkpoint rows, 77 satisfy $t_{\text{IR}} = 0.3$, and all 77 also satisfy $t_{\text{SR}} = 0.95$. Thus IR locates reduced same-history sensitivity in this regime, while SR confirms that the tested

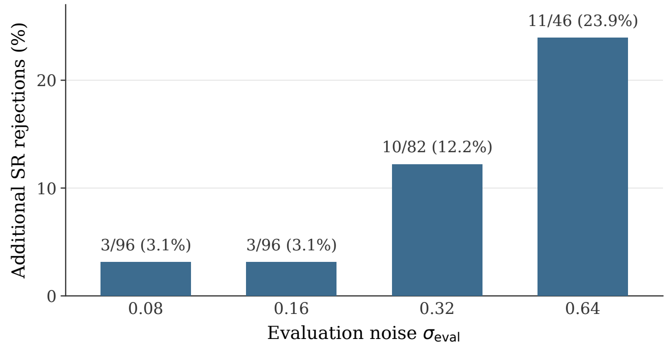

Figure 3: Percentage of relative-IR-passing checkpoints rejected by the fixed SR condition as evaluation noise increases. Annotations report rejection counts and conditional denominators. Thresholds and state pairs are fixed.

state-coordinate distinctions remain separated. The rise in SR mainly follows contraction of the same-history radius; median different-state distances remain largely stable.

4.3 SR Becomes Active at Higher Severity

Does SR begin to affect the screen when evaluation noise grows? We keep the checkpoint set, rollout protocol, pair catalog, normalization, and thresholds fixed. Relative IR compares each radius with a severity-matched unaugmented reference; SR compares that radius with clean distances between tested state pairs. Common growth can cancel in relative IR while crossing the fixed state-separation scale, so the conditions need not agree.

Among IR-passing rows, SR rejects 3/96 checkpoints at both $\sigma_{\text{eval}} = 0.08$ and 0.16, rising to 10/82 and 11/46 at the two harder severities. Some harder-severity rows instead pass SR but fail IR, confirming that the conditions are non-equivalent. Lower IR and higher SR are also positively associated on average with retention of planning performance after accounting for training severity, evaluation severity, and their ratio, although effects vary across tasks (Appendix D). These results show activation of the separation guard, not superior checkpoint selection.

4.4 ACPC and Prediction-Error Change

We predict the bounded error drift $d = |e_{\tilde{h}} - e_h|$ at $H = 8$ using each unaugmented task–run checkpoint, noise $\sigma \in \{0.02, 0.05, 0.08\}$, and two draws. Sixteen-fold group cross-validation keeps every trajectory group intact.

All ridge models contain severity, encoder shift, and one-step ACPC. The same-horizon control adds the best test-performing eight-step variant with zeroed, cross-trajectory, or temporally shuffled actions; the ACPC model adds the recorded-action value. This isolates the correct action sequence from rollout length. Appendix E gives full controls.

Recorded-action ACPC$_8$ is best in all 12 task–run cells (Figure 4). Averaging tasks within each run, it lowers MAE by $55.9 \pm 4.7\%$ relative to Base and $51.3 \pm 3.5\%$ relative to the strongest destroyed-action control (mean $\pm$ sample SD across three runs). This concerns error change, not absolute prediction accuracy.

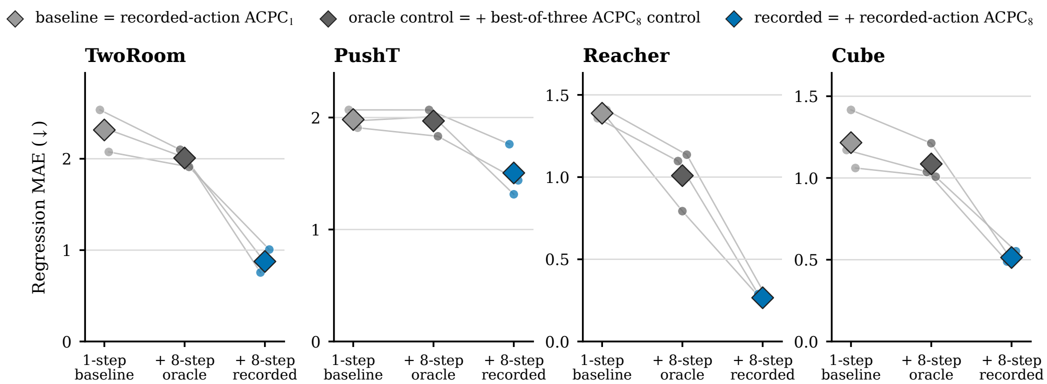

Figure 4: Group-cross-validated MAE for prediction-error drift; lower is better. Points are training runs and diamonds are task means. Recorded-action ACPC$_8$ is best in all 12 task–run cells. Scales are task-specific.

4.5 ACPC and the Cost of CEM Plan Changes

A perturbation can make CEM choose a different plan. We measure its extra cost under the clean-history model, $(C_h(\tilde{\mathbf{a}}) - C_h(\mathbf{a}))_+$, and call this single-decision quantity *CEM selection regret*. It is model cost, not simulator return.

Paired CEM runs share the initial proposal and random numbers but update from their own elites. We regress $\log(1 + \text{selection regret})$ across held-out tasks. The baseline contains severity, training condition, one-step candidate-ACPC q90, and the clean best–second-best cost gap; the expanded model adds five-step candidate-ACPC q90, matching the planning horizon. Appendix I gives the budget and samples.

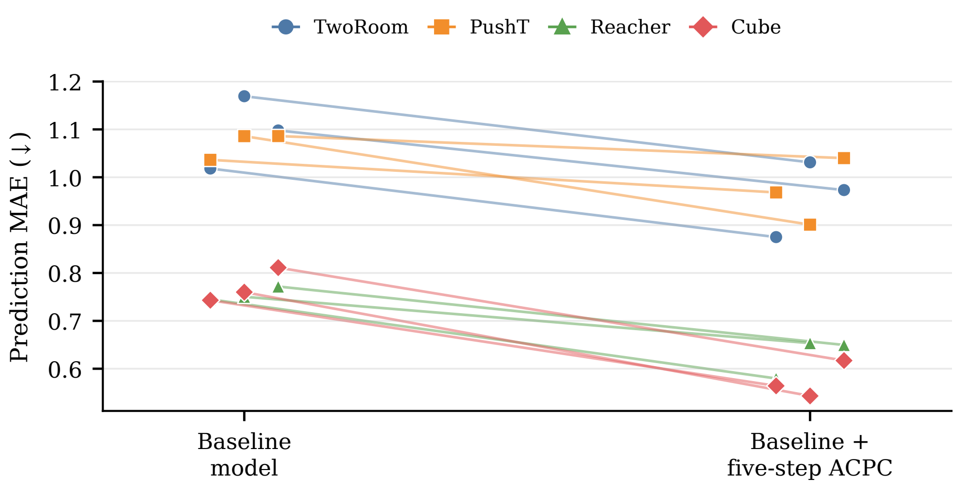

Figure 5: Planner-horizon ACPC improves prediction of CEM selection regret. MAE is computed for $\log(1 + \text{selection regret})$ and decreases in all 12 paired task–run comparisons.

Five-step ACPC reduces MAE in all 12 task–run evaluations. Averaging task MAEs equally within each training run and then computing the per-run reduction gives a mean reduction of 15.2% (a 2.0 percentage-point sample SD across the three runs; Figure 5). On fixed first-round CEM candidate pools, the top-1 sufficient condition covers 34.8–69.5% of histories for augmented checkpoints across probe severities, versus 0.0–2.3% for unaugmented checkpoints. Appendix I reports the full fixed-pool analysis; it does not certify later adaptive rounds.

4.6 CHECKPOINT SCREENING ACROSS TASKS

Across 14 source/test task partitions, 13 choose $(t_{\mathrm{IR}}, t_{\mathrm{SR}}) = (0.3, 0.95)$; Reacher alone chooses $(0.1, 0.95)$. With two or three source tasks, the first accepted level lies within 0.5 grid steps of recovery on average and never differs by more than one (balanced accuracy 0.900; precision/recall $0.913/0.953$). Single-source selection is less stable (mean balanced accuracy 0.855), with Reacher delaying acceptance by up to five levels. Full partitions are in Appendix F. Both values in the predominant $(0.3, 0.95)$ threshold pair lie at the upper ends of their tested grids, so sensitivity beyond them remains unresolved.

4.7 DIAGNOSTIC BEHAVIOR ON PLDM

We apply the same definitions to the complete PLDM training sweep. For every task, all eight augmented-level three-run means lie above the unaugmented mean in noisy-observation planning success and SR, while relative IR lies below its baseline of one (Table 1). This is a descriptive model-family check, not a recovery classification or a transfer of LeWM thresholds.

Table 1: **PLDM diagnostics across the augmented training grid.** Values are three-run means; brackets give their range across all eight augmented levels. $\pm$ denotes the unaugmented across-run population SD, and unaugmented relative IR equals 1 by definition.

Task	Planning success (%) unaug. $\to$ augmented range	Relative IR augmented range	SR unaug. $\to$ augmented range
TwoRoom	67.1±8.0→[95.7, 97.7]	[0.071, 0.440]	0.082→[0.967, 1.000]
PushT	10.2±5.9→[20.9, 70.3]	[0.070, 0.866]	0.228→[0.439, 1.000]
Reacher	53.0±29.7→[63.0, 82.2]	[0.166, 0.893]	0.650→[0.887, 1.000]
Cube	45.9±1.7→[50.6, 53.9]	[0.043, 0.742]	0.170→[0.637, 1.000]

4.8 CHECKPOINT COMPARISON UNDER BLUR AND RESIZE

We compare unaugmented and $\sigma_{\max} = 0.08$ checkpoints under blur ($k = 15$) and resize (scale 0.25), yielding 24 task–run–shift pairs. IR improvement ($1 - \mathrm{IR}^{\mathrm{rel}}$) and SR change each agree with the prespecified behavioral label in 22/24 pairs. Their pooled Spearman associations with planning-success change are 0.854 and 0.913, respectively, and remain positive in every leave-one-task-out subset, although task-level relations are heterogeneous. SR vetoes none of the six endpoint checkpoints passing IR. Thus the components retain descriptive signal for these shifts, but this experiment does not show incremental screening utility from SR; Appendix G gives per-task results and uncertainty analyses.

5 DISCUSSION AND LIMITATIONS

Multi-step action-conditioned comparison exposes perturbation effects that encoder-only and one-step measurements can miss. At checkpoint level, IR and SR answer different questions: IR measures same-history sensitivity, whereas SR asks whether tested state distinctions survive that contraction. Their regime-dependent disagreement supports reporting them separately rather than collapsing them into one continuous score. A PushT case study visualizes this contraction relative to the spacing between clean histories, both at the encoder output and after rollout (Appendix B).

The scope is deliberately narrower than a robustness certificate. The screen requires a matched unaugmented checkpoint, observed future frames for IR normalization, and dataset labels for SR pairs; nominal clean quality remains a separate requirement. The cost and selection conditions apply to evaluated candidate pools and do not guarantee later adaptive CEM rounds or environment return. Blur and resize cover one severity each, and the state-pair catalog cannot test distinctions it does not encode.

6 Conclusion

ACPC measures how a visual perturbation propagates through an action-conditioned latent rollout. We derive bounds on prediction-error and fixed-pool planning-cost changes and validate the metric at pair and planner horizons. Across checkpoints, IR measures perturbation sensitivity and SR guards tested state separation, becoming more active at higher evaluation severity. Together they form a scoped offline diagnostic, not a universal performance or planning certificate.

AI use statement

Generative AI tools were used to assist with code generation and debugging, language editing, and auxiliary checks of experimental results. The authors specified the research questions, methods, and experimental protocols; independently verified all AI-assisted code, proofs, numerical results, and manuscript claims; and take full responsibility for the submission.

Reproducibility statement

The appendix specifies tasks, perturbations, checkpoints, seeds, sampling, normalization, thresholds, and aggregation. The anonymous supplementary material includes the code and machine-readable results needed to reproduce the reported figures and tables.

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A Proofs and Fixed-Pool Analysis

Proof of Proposition 1. Let $u = \bar{G}_{\mathbf{a}}(E_\theta(h))$, $v = \bar{G}_{\mathbf{a}}(E_\theta(\tilde{h}))$, and $y = Y^H_{\mathbf{a}}$. Both errors use the same target y in the weighted rollout space of Equation (2). The reverse triangle inequality gives

$$|e_{\tilde{h}} - e_h| = \big|\, \|v-y\|_2 - \|u-y\|_2 \,\big| \tag{10}$$

$$\le \|(v-y)-(u-y)\|_2 = \|v-u\|_2 = \mathrm{ACPC}_H(h,\tilde{h},\mathbf{a}). \tag{11}$$

The one-sided error increase satisfies the same bound because $(e_{\tilde{h}} - e_h)_+ \le |e_{\tilde{h}} - e_h|$.

The following result uses the cost-change bounds for individual candidates to determine whether the selected candidate or elite set can change.

Proposition 3 (Fixed-pool selection certificates)**.** *In the setting of Proposition 2, let w be the unique winner for the clean input. If its cost advantage over every other candidate exceeds the combined cost-change bounds,*

$$\min_{j\neq w}\big(C_j - C_w - b_j - b_w\big) > 0, \tag{12}$$

then it remains the winner for the perturbed input. Similarly, let $\mathcal{E}$ be the clean top-k elite set. If

$$\min_{i\in\mathcal{E},\, j\notin\mathcal{E}}\big(C_j - C_i - b_j - b_i\big) > 0, \tag{13}$$

then the perturbed input produces the same elite set.

Corollary 1 (Conditional CEM stability)**.** *Consider clean and perturbed CEM runs that start from the same proposal and use the same random samples. Suppose that, at a given iteration, they evaluate corresponding candidate action sequences. If Equation* (13) *holds, they select the same elite candidates and fit the same proposal for the next iteration. If the condition holds at every iteration, the two runs remain aligned. Because the evaluated solver returns the final proposal mean as its action, the selected actions are also identical.*

Proof of Proposition 2. For a single candidate, omit the index. The difference between the two squared costs factors as

$$|\tilde{C} - C| = \big|\|\tilde{x}-g\|_2^2 - \|x-g\|_2^2\big| \tag{14}$$

$$= \big|\langle \tilde{x}-x, \tilde{x}+x-2g\rangle\big| \tag{15}$$

$$\le \|\tilde{x}-x\|_2\, \|\tilde{x}+x-2g\|_2 \tag{16}$$

$$\le r\big(\|x-g\|_2 + \|\tilde{x}-g\|_2\big) = b. \tag{17}$$

The first inequality follows from Cauchy–Schwarz, and the second follows from the triangle inequality.

Let w and $\tilde{w}$ be the winners for the clean and perturbed inputs. Since $\tilde{w}$ minimizes the perturbed cost,

$$C_{\tilde{w}} \le \tilde{C}_{\tilde{w}} + b_{\tilde{w}} \le \tilde{C}_w + b_{\tilde{w}} \le C_w + b_w + b_{\tilde{w}},$$

Since w minimizes the clean cost, $C_{\tilde{w}} - C_w \ge 0$. Together, these inequalities prove Equation (5).

For any clean winner w and competitor j,

$$\tilde{C}_j - \tilde{C}_w \ge C_j - C_w - |\tilde{C}_j - C_j| - |\tilde{C}_w - C_w| \ge C_j - C_w - b_j - b_w.$$

Therefore, Equation (12) ensures that every competitor remains more costly than w. Applying the same argument to every $i \in \mathcal{E}$ and $j \notin \mathcal{E}$ proves that Equation (13) preserves the elite set. The strict inequalities exclude ties and make the result independent of the tie-breaking rule.

Proof of Corollary 1. Index the clean and perturbed runs by $b \in \{\mathrm{c}, \mathrm{p}\}$ and the CEM iterations by t. Let ϕ_t^b denote the proposal mean and variance for run b. Given shared random samples ω_t, the deterministic sampling map T produces the candidate pool

$$\mathcal{A}_t^b = T(\phi_t^b, \omega_t).$$

Each run selects the indices $\mathcal{E}_t^b$ of its k lowest-cost candidates. A deterministic, permutation-invariant update U then fits the next proposal:

$$\phi_{t+1}^b = U(\{\mathbf{a}^j : j \in \mathcal{E}_t^b\}).$$

The evaluated solver satisfies this condition because it uses the unweighted mean and standard deviation of the elite candidates.

The two runs start from the same proposal, so $\phi_0^{\mathrm{c}} = \phi_0^{\mathrm{p}}$. If their proposals agree at iteration t, the shared samples generate the same candidate pool. When Equation (13) holds, the two runs also select the same elite candidates and therefore fit the same next proposal:

$$\phi_t^{\mathrm{c}} = \phi_t^{\mathrm{p}} \implies \mathcal{A}_t^{\mathrm{c}} = \mathcal{A}_t^{\mathrm{p}} \implies \mathcal{E}_t^{\mathrm{c}} = \mathcal{E}_t^{\mathrm{p}} \implies \phi_{t+1}^{\mathrm{c}} = \phi_{t+1}^{\mathrm{p}}.$$

Thus, if the condition holds at every iteration, the proposals and candidate pools remain identical. The evaluated solver returns the final proposal mean, so the selected actions are also identical.

If a solver instead returns the best candidate from the final pool, Equation (12) must also hold at the last iteration. If either certificate cannot be verified, the proof no longer guarantees that the subsequent elite sets, proposals, or candidate pools agree.

Relation to ACPC. The planning-cost bound uses the final-step displacement r_j, whereas ACPC combines displacements across all H rollout steps. From Equation (2),

$$\sqrt{\alpha_H}\, r_j \leq \mathrm{ACPC}_H \qquad \text{when } \alpha_H > 0.$$

Thus, $r_j \leq \mathrm{ACPC}_H / \sqrt{\alpha_H}$. In the planner experiments, we compute r_j directly instead of using this upper bound. This gives a tighter candidate-level measurement and does not require the observed future, but it requires rolling out each candidate from both the clean and perturbed histories.

Tightness of the bounds. Both bounds can hold with equality, so their right-hand sides cannot be reduced without additional assumptions. For the prediction-error bound, let u be a unit vector, set the common target to $Y = 0$, and let the two predicted rollouts be su and tu, where $s, t \geq 0$. Then

$$\big|\, \|tu\|_2 - \|su\|_2 \,\big| = \|tu - su\|_2,$$

so Equation (3) holds with equality. For the planning-cost bound, set $g = 0$, $x = su$, and $\tilde{x} = tu$. Then

$$|\tilde{C} - C| = |t - s|(t + s) = r\big(\|x - g\|_2 + \|\tilde{x} - g\|_2\big),$$

so the candidate cost-change bound is also attained exactly.

B Local-Geometry Case-Study Definitions

This supplementary case study visualizes how augmentation contracts perturbed views relative to nearby clean histories. We use 128 logged PushT histories. For each history, we generate one clean view and 18 perturbed views, with six draws at each standard deviation in $\{0.01, 0.04, 0.08\}$. Each of the four qualitative t-SNE panels uses a separate deterministic projection (van der Maaten & Hinton, 2008), so coordinates are not compared across panels.

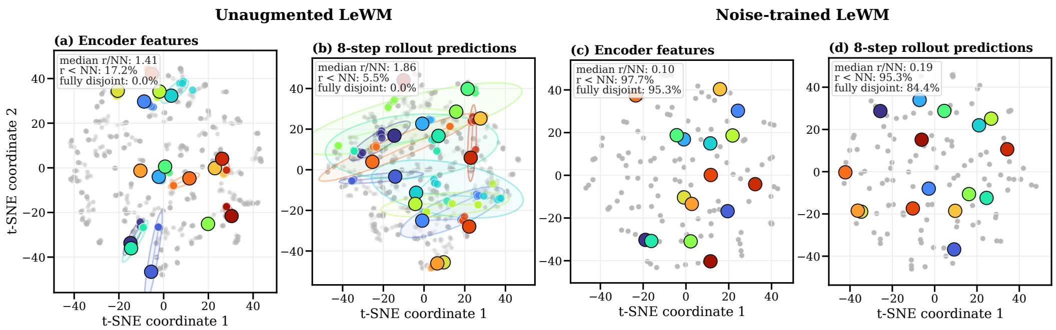

Figure 6: Local geometry for matched unaugmented (left) and $\sigma_{\max} = 0.08$ PushT checkpoints (right), at the encoder and after eight rollout steps. Markers show clean histories and perturbed views; summaries are computed in the original 192-dimensional space, not in t-SNE.

Without augmentation, median r/NN is 1.41 at the encoder and 1.86 after rollout, with no fully disjoint perturbation clouds. With augmentation, the ratios fall to 0.10 and 0.19, and the disjoint fractions rise to 95.3% and 84.4%. This is descriptive geometry, not evidence of task-semantic separation or planning robustness.

We evaluate two representation spaces: the encoder output, denoted by $\mathcal{V} = \mathrm{enc}$, and the representation after eight autoregressive rollout steps, denoted by $\mathcal{V} = \mathrm{roll}$. The rollout horizon is $H = 8$ (Section 3.1). For history i, let $v_{i,\mathcal{V}}^{(0)}$ be its clean representation and let $v_{i,\mathcal{V}}^{(m)}$, $m = 1, \ldots, M$ with $M = 18$, be its perturbed representations. In the rollout space, the clean and perturbed views of the same history use the same recorded action sequence. We then define the sampled visual radius, the distance to the nearest other clean history, and their ratio:

$$r_{i,\mathcal{V}}^{\mathrm{vis}} = \max_{1 \le m \le M} \left\| v_{i,\mathcal{V}}^{(m)} - v_{i,\mathcal{V}}^{(0)} \right\|_2 ,$$

$$d_{i,\mathcal{V}}^{\mathrm{NN}} = \min_{j \ne i} \left\| v_{i,\mathcal{V}}^{(0)} - v_{j,\mathcal{V}}^{(0)} \right\|_2 , \qquad \rho_{i,\mathcal{V}}^{\mathrm{local}} = \frac{r_{i,\mathcal{V}}^{\mathrm{vis}}}{d_{i,\mathcal{V}}^{\mathrm{NN}}}. \tag{18}$$

The figures abbreviate $\rho_{i,\mathcal{V}}^{\mathrm{local}}$ as r/NN. A value below one means that every sampled perturbation moves the representation by less than the distance from its clean anchor to any other clean anchor. A value of at least one means that at least one perturbation displacement is as large as the nearest-clean distance. This comparison uses only distance magnitudes: it does not show that a perturbed representation approaches or overlaps another history, and the nearest clean history need not differ across a task-relevant state boundary.

The r/NN ratio considers only the perturbation radius of anchor i. To account for the perturbation radii of both histories, we also compare their enclosing balls. The ball around anchor i is *fully disjoint* if it does not intersect the ball around any other anchor:

$$\left\| v_{i,\mathcal{V}}^{(0)} - v_{j,\mathcal{V}}^{(0)} \right\|_2 > r_{i,\mathcal{V}}^{\mathrm{vis}} + r_{j,\mathcal{V}}^{\mathrm{vis}} \quad \text{for every } j \ne i. \tag{19}$$

Across the 128 anchors, we report three summaries: the median r/NN ratio, the fraction with $r_{i,\mathcal{V}}^{\mathrm{vis}} < d_{i,\mathcal{V}}^{\mathrm{NN}}$, and the fraction satisfying the fully disjoint condition in Equation (19). The second summary compares each perturbation radius with the distance to the nearest other clean anchor. The third also accounts for the perturbation radius around every other anchor. All three summaries are computed in the original high-dimensional representation; overlap between the qualitative t-SNE ellipses is not used.

These summaries are descriptive and depend on the sampled histories and perturbation draws. They compare the encoder output and the final rollout step but do not examine intermediate predictions. They also do not bound changes in prediction error or planning cost; those theoretical links are provided by pairwise ACPC in Section 3.1.

C Evaluation and Diagnostic Protocol

This appendix specifies the fixed evaluation and diagnostic protocol used in the main tables.

Evaluation grid. We evaluate LeWM on PushT, TwoRoom, Reacher, and Cube. Each checkpoint is evaluated with seeds 42, 43, and 44, using 100 episodes per seed. The primary evaluation adds Gaussian noise with standard deviation 0.08 to the history observations while keeping the goal image clean. For each task, the training grid contains one unaugmented checkpoint and eight checkpoints trained with full-sequence Gaussian-noise augmentation at $\sigma_{\max} \in \{0.01, 0.02, \ldots, 0.08\}$.

PLDM protocol. PLDM is evaluated on the same four tasks and nine-checkpoint training grid as LeWM, using the same evaluation seeds and number of logged histories. Its diagnostics use the same clean–perturbed pairing, eight-step rollout horizon, and SR definition. In the PLDM plots, raw IR is divided by the value of the unaugmented PLDM checkpoint from the same task and training run.

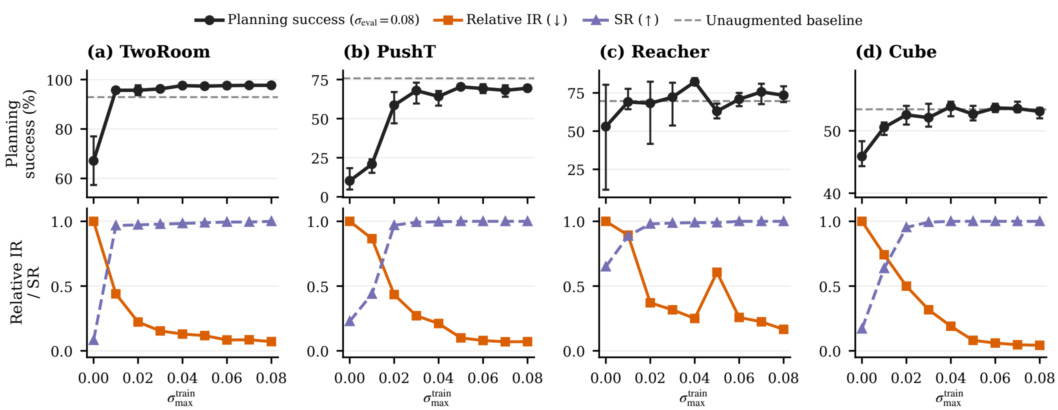

Figure 7: Complete PLDM Gaussian-noise training sweep over three runs. Curves show three-run means and success bars span the across-run range. The main-text summary includes every augmented level rather than selecting a recovery subset.

Success-rate criterion. For each task and training run, let P_{base} be the noisy-observation success rate of the unaugmented checkpoint, and let P_{upper} be the highest noisy-observation success rate observed in the fixed nine-checkpoint sweep. A checkpoint meets the success-rate criterion if its noisy-observation success rate is at least

$$P_{\text{base}} + 0.8(P_{\text{upper}} - P_{\text{base}})$$

and its clean-observation success rate is no more than five percentage points below that of the unaugmented checkpoint. At the task level, we call an augmentation level recovered only when this criterion holds in at least two of the three training runs. P_{upper} is used only to scale this recovery label; every level meeting the criterion remains part of the reported range.

IR protocol. We compute IR separately for each task, training run, and checkpoint. To express rollout divergence relative to the normal motion in each history, we first compute a clean-motion scale. For anchor i, let $u_{i,0}^{\text{obs}}$ be the projected embedding of the final history observation, and let $u_{i,1:H}^{\text{obs}}$ be the projected embeddings of the next H observed clean frames. We define

$$s_i = Q_{0.50}\Big(\big\{\|u_{i,k}^{\text{obs}} - u_{i,k-1}^{\text{obs}}\|_2\big\}_{k=1}^{H}\Big). \tag{20}$$

Thus, s_i is the median displacement between consecutive clean observations, including the transition from the final history observation to the first future observation.

For each anchor, we generate multiple perturbation draws using the visual shift being evaluated. The Gaussian-noise experiments use $\sigma = 0.08$; the blur and resize experiments apply their corresponding shifts. For every draw, we compute eight-step ACPC with uniform weights $\alpha_k = 1/H$ and normalize it by $s_i + \varepsilon_{\text{norm}}$. We average the normalized values across draws for each anchor and then take q90 across anchors. The result is raw IR in Equation (7).

SR compares different-state distances with raw IR from the same checkpoint. For the sweep plots, raw IR is divided by the value of the corresponding unaugmented checkpoint, as defined in Equation (9). Only the LeWM cross-task analysis uses relative IR for threshold selection. We use $\varepsilon_{\text{norm}} = 10^{-8}$ in Equation (6). Table 2 reports the results for the tested rollout horizons and summary quantiles.

Table 2: Horizon and quantile sensitivity of the IR reduction at fixed evaluation noise $\sigma = 0.08$. Each entry is the median, over the three training runs, of the ratio between the IR of the checkpoint trained at the highest augmentation level ($\sigma_{\max} = 0.08$) and that of the unaugmented checkpoint; lower means a larger reduction. These values are descriptive and do not retune any threshold.

	$q = 0.90$ by horizon				$H = 8$ by quantile		
Task	H1	H2	H4	H8	q80	q90	q95
TwoRoom	0.05	0.04	0.05	0.06	0.06	0.06	0.06
PushT	0.06	0.07	0.06	0.09	0.07	0.09	0.09
Reacher	0.02	0.03	0.03	0.02	0.02	0.02	0.02
Cube	0.04	0.03	0.04	0.04	0.03	0.04	0.04

SR protocol. SR follows Equation (8). We first define a local neighborhood using the standardized logged endpoint states. Let $d_{0.35}$ be the 35th percentile of all pairwise distances between different endpoints—that is, the q35 of all off-diagonal Euclidean distances. This cutoff keeps the comparisons local while retaining an eligible neighbor for most anchors; Table 3 reports the resulting counts.

For each anchor i, we select the nearest history $j(i)$ that has a different endpoint-state label and lies within distance $d_{0.35}$. If no such history exists, the anchor is excluded from SR. We roll both histories forward under the anchor's recorded action sequence $\mathbf{a}_i$. This sequence was not generally executed by the neighboring history, but using it for both rollouts prevents action differences from affecting the comparison. Their normalized rollout distance is

$$D_i^{\text{diff}} = \frac{\left\|\bar{G}_{\mathbf{a}_i}(E_\theta(h_i)) - \bar{G}_{\mathbf{a}_i}(E_\theta(h_{j(i)}))\right\|_2}{s_i + \varepsilon_{\text{norm}}}. \tag{21}$$

The pair counts as separated when D_i^{diff} is greater than raw q90 IR plus the fixed margin 0.10.

For pairing, the endpoint state is taken after the eight observed future steps and therefore records the state reached under that trajectory's own actions. The dataset preprocessor standardizes each state coordinate, and neighborhood distances use the full standardized state vector. Endpoint-state labels are constructed by median splits of the task-specific coordinates listed in Table 3, using the fixed batch of 100 endpoints generated with anchor seed 9101. Because the logged endpoints, labels, and neighborhoods are fixed before checkpoint evaluation, the selected pairs and eligible-anchor counts are identical across checkpoints and training runs.

Table 3: Endpoint-state labels used to construct SR pairs. The state is taken at the eighth observed future step, and its coordinates are standardized using full-dataset statistics. Counts report eligible and skipped histories among the fixed 100 anchors. These labels define operational partitions for the diagnostic; they are not semantic or action-relevance annotations.

Task	Logged state field (dim.)	Label construction $\ell(y)$	Eligible / skipped
TwoRoom	`pos_agent` (2)	Median split of the standardized agent x coordinate (`pos_agent[0:1]`).	61 / 39
PushT	`state` (7)	Three-bit label from median splits of standardized block x, block y, and block angle (`state[2:5]`).	98 / 2
Reacher	`observation` (6)	Three-bit label from median splits of the two standardized joint velocities and their Euclidean norm (`observation[4:6]`).	100 / 0
Cube	`observation` (28)	Three-bit label from median splits of the first two standardized coordinates and $\|r\|_2$, where $r = \bar{o}_{0:3} - \bar{o}_{25:28}$.	100 / 0

Threshold grids. For the LeWM cross-task evaluation, we search

$$t_{\mathrm{IR}} \in \{0.05, 0.075, 0.10, 0.15, 0.20, 0.30\}, \qquad t_{\mathrm{SR}} \in \{0.80, 0.85, 0.90, 0.95\}.$$

For each nonempty subset of tasks used for threshold selection, we evaluate every threshold pair. We first minimize the mean number of augmentation-grid steps between the first checkpoint accepted by the IR–SR screen and the first checkpoint that meets the success-rate criterion. Remaining ties are resolved by, in order, fewer early acceptances, fewer late acceptances, higher balanced accuracy, a smaller t_{IR}, and a larger t_{SR}.

The selected thresholds are then applied unchanged to the tasks that were not used for their selection. Success-rate labels from those tasks and all blur or resize results are excluded from threshold selection.

Reproducibility. The commands used to rebuild the paper figures and tables are documented in `paper1/scripts/README.md` and `tools/README_paper1.md`. The released machine-readable summaries are sufficient for the reader-facing aggregation and plotting steps. Recomputing checkpoint-level diagnostics additionally requires the released checkpoints and datasets. The summaries record the training seeds, evaluation seeds, and protocol hashes used for each result.

Table 4: Recovery ranges in the original Gaussian-noise training sweep (nine checkpoints per task: no augmentation plus eight noise levels; Figure 2 shows every level). The final column lists the contiguous training-noise ranges that meet the prespecified success-rate criterion in Section 4.1. It summarizes an accepted range rather than selecting a single training strength.

Task	Unaugmented success (%)	Training levels meeting criterion
TwoRoom	68.8	0.01–0.08
PushT	7.2	0.03–0.08
Reacher	18.2	0.03–0.08
Cube	43.1	0.03–0.07

D Additional Multi-Severity Analysis

This section gives the aggregation and sensitivity details behind Section 4.3. The broad-severity experiment is analyzed separately from the original narrow sweep. It keeps the original thresholds and pair construction fixed and uses continuous retention rather than defining a new performance band.

The same 3 TwoRoom checkpoints account for every additional SR rejection among IR-passing rows at $\sigma_{\mathrm{eval}} = 0.08$ and 0.16: 1 has $\sigma_{\max} = 1.0$ and 2 have $\sigma_{\max} = 1.5$. At $\sigma_{\mathrm{eval}} = 0.32$, these rejections comprise 4 TwoRoom, 5 PushT, and 1 Reacher checkpoints; at 0.64, they comprise 5 TwoRoom and 6 PushT checkpoints. No Cube checkpoint passes IR while failing SR. The conditional rejections thus span three tasks at 0.32 and two at 0.64, although some checkpoint rows recur across severity slices and the conditional denominator also changes. All low-severity rejections fall on TwoRoom, whose SR catalog uses one median split and has the fewest eligible anchors (61 of 100; Table 3); this concentration should be read within the scope of the tested pair catalog.

Table 5: Fixed-threshold decision decomposition in the separate broad-severity experiment. Each row partitions the same 96 trained checkpoints at one evaluation severity. “IR pass, SR fail” is the additional rejection shown in Figure 3. The reverse disagreement appears at the two higher severities, so neither condition subsumes the other.

σ_{eval}	Pass both	IR pass, SR fail	IR fail, SR pass	Fail both
0.08	93	3	0	0
0.16	93	3	0	0
0.32	72	10	5	9
0.64	35	11	5	45

The reverse disagreement in Table 5 is expected to be possible because the conditions use different references. Relative IR divides the raw perturbation radius by the unaugmented checkpoint’s radius at the same severity. SR instead compares that raw radius with clean different-state distances from the checkpoint under evaluation. The fixed-threshold results therefore cannot be reproduced by nesting one condition inside the other.

For each task and training run, the grid contains eight training-noise maxima and four evaluation severities. Let x_{ij} denote any diagnostic or behavioral quantity at training level i and evaluation severity j. We remove the two additive main effects using

$$\widetilde{x}_{ij} = x_{ij} - \overline{x}_{i\cdot} - \overline{x}_{\cdot j} + \overline{x}_{\cdot\cdot}.$$

Spearman correlations are computed over the 32 centered cells within a single task and training run. We then average the 3 training-run correlations within each task and weight the 4 tasks equally. The task ranges in Table 6 summarize heterogeneity; they are not confidence intervals. Double-centering leaves $(8-1)(4-1) = 21$ interaction degrees of freedom and imposes row- and column-sum constraints, so the 32 cells are not treated as independent observations.

Table 6: Descriptive Spearman associations with same-checkpoint retention (success under noise minus clean success) in the separate broad-severity experiment. Within each task and training run, all quantities are double-centered over the 8×4 training–evaluation severity grid. Training-run correlations are then averaged within each task, and tasks receive equal weight. The adjusted column reports partial Spearman correlation after residualizing the rank-transformed diagnostic and outcome against the rank-transformed exposure ratio $-\sigma_{\mathrm{eval}}/\sigma_{\max}$ within each task–run block. Ranges contain the four task means, not confidence intervals. The exposure ratio is a simple metadata comparator, not an exhaustive baseline, and these associations do not establish checkpoint-selection accuracy.

Quantity (favorable direction)	Direct ρ	Task range	Adjusted ρ	Task range
$-\sigma_{\mathrm{eval}}/\sigma_{\max}$	0.787	[0.662, 0.887]	–	–
$-\mathrm{IR}^{\mathrm{rel}}$	0.796	[0.671, 0.909]	0.458	[0.070, 0.667]
SR	0.826	[0.756, 0.900]	0.532	[0.225, 0.694]

On PushT, the exposure comparator's direct task-level correlation exceeds both diagnostics. This is why we use it as an explicit design-metadata comparator and do not interpret the small differences among the direct aggregate values as diagnostic superiority.

Using stress success divided by same-checkpoint clean success instead of their percentage-point difference gives the same qualitative result. The equal-task mean direct correlations are 0.813 for lower relative IR and 0.839 for higher SR; after accounting for the exposure ratio they are 0.560 and 0.626, respectively. Every clean-success estimate is nonzero (minimum 7%), but ratios can still amplify changes for weak checkpoints; we therefore keep the percentage-point difference as the primary outcome and treat the ratio only as a sensitivity check. These are descriptive factorial associations. The 384 rows are not independent replicates, and the four evaluation-severity rows for a checkpoint share the same clean-performance estimate; therefore we do not attach row-level p-values or confidence intervals.

E Prediction-Error Scope and Controls

We checked Proposition 1 for every logged eight-step pair and every candidate-specific five-step pair across all tasks, training runs, and training conditions. No numerical violations occurred. This check confirms that the implementation follows the inequality. It is not an independent statistical success count or evidence of model quality, because the inequality holds for every correctly computed pair.

The prediction-error and planning experiments answer different questions. The prediction-error experiment measures how a visual perturbation changes rollout error relative to the observed future under the recorded actions. The planning experiment instead computes ACPC for the action candidates evaluated by CEM and relates it to selection regret. We therefore treat the planning analysis as separate evidence rather than infer it from the prediction-error result (Section 4.5).

The prediction-error analysis uses only the unaugmented checkpoints and Gaussian-noise levels $\sigma \in \{0.02, 0.05, 0.08\}$. Table 7 reports the relative reduction in cross-validated MAE for each training run. Table 8 reports the corresponding MAE values and the selected destroyed-action controls. Base+ACPC$_8$ achieves lower MAE than both comparison models on all four tasks in every training run.

Table 7: Relative reduction in out-of-group MAE from Base+ACPC$_8$ when predicting the error drift d on the unaugmented checkpoints (protocol and model definitions in Section 4.4). Base+Control$_8$ uses the per-cell oracle control; higher is better. The last column counts task–run cells in which Base+ACPC$_8$ is best.

Training run (seed)	vs. Base	vs. Base+Control$_8$	Task–run cells improved
3072	55.5%	51.4%	4/4
3073	60.8%	54.8%	4/4
3074	51.4%	47.7%	4/4
mean ± SD	55.9 ± 4.7%	51.3 ± 3.5%	12/12

Table 8: Out-of-group MAE (16-fold leave-one-trajectory-group-out) for predicting the error drift d on the unaugmented checkpoints; lower is better, model definitions in Section 4.4. "Selected control" is the destroyed-action control chosen by the per-cell oracle; "paired wins" counts the folds (of 16) in which Base+ACPC$_8$ beats Base+Control$_8$.

Task	run (seed)	Base	Base +Control$_8$	Base +ACPC$_8$	selected control	paired wins
TwoRoom	3072	2.535	2.099	0.755	shuffled actions	15/16
TwoRoom	3073	2.336	2.015	0.866	shuffled actions	15/16
TwoRoom	3074	2.075	1.911	1.006	swapped actions	13/16
PushT	3072	2.068	2.068	1.762	shuffled actions	11/16
PushT	3073	1.969	2.008	1.315	zeroed actions	13/16
PushT	3074	1.909	1.833	1.439	zeroed actions	13/16
Reacher	3072	1.358	1.097	0.288	shuffled actions	16/16
Reacher	3073	1.399	0.793	0.247	shuffled actions	16/16
Reacher	3074	1.409	1.136	0.263	shuffled actions	16/16
Cube	3072	1.171	1.037	0.489	zeroed actions	14/16
Cube	3073	1.416	1.212	0.500	zeroed actions	14/16
Cube	3074	1.061	1.008	0.552	shuffled actions	14/16

F Cross-Task Threshold Partitions

With four tasks, there are 14 nonempty choices of tasks for threshold selection: four single-task choices, six two-task choices, and four three-task choices. The selected thresholds are applied to the remaining task or tasks. Each evaluation includes all three training runs and all nine checkpoints for every remaining task. Checkpoints are not counted as independent samples: metrics first average the training runs within each evaluation task and then weight the evaluation tasks equally.

When thresholds are selected using only Reacher, the procedure chooses $(t_{\mathrm{IR}}, t_{\mathrm{SR}}) = (0.1, 0.95)$ instead of the predominant $(0.3, 0.95)$. Both threshold pairs locate Reacher recovery within one augmentation-grid step, but $t_{\mathrm{IR}} = 0.1$ matches the Reacher onset exactly and therefore wins the tie-break. Recovered Reacher checkpoints have relative IR well below 0.1, whereas the earliest recovered checkpoints on the other tasks remain above 0.1. The Reacher-only threshold therefore accepts recovery on those tasks too late, by as many as five augmentation levels.

Table 9: Joint IR/SR checkpoint screening with thresholds chosen on other tasks. Thresholds are selected on the threshold-selection tasks and applied unchanged to all test tasks. Metrics first average training runs within each test task and then weight tasks equally. Recovery-onset mismatch is measured in training-augmentation grid levels (one level is 0.01 in $\sigma_{\max}$).

Selection tasks	Test tasks	t_{IR}	t_{SR}	BA	P / R	Onset mismatch mean / max
TwoRoom	PushT, Reacher, Cube	0.3	0.95	0.888	0.884 / 0.981	0.3 / 1.0
PushT	TwoRoom, Reacher, Cube	0.3	0.95	0.894	0.902 / 0.956	0.6 / 1.0
Reacher	TwoRoom, PushT, Cube	0.1	0.95	0.723	0.906 / 0.540	2.9 / 5.0
Cube	TwoRoom, PushT, Reacher	0.3	0.95	0.913	0.950 / 0.938	0.6 / 1.0
TwoRoom, PushT	Reacher, Cube	0.3	0.95	0.874	0.853 / 1.000	0.3 / 1.0
TwoRoom, Reacher	PushT, Cube	0.3	0.95	0.887	0.873 / 0.972	0.3 / 1.0
TwoRoom, Cube	PushT, Reacher	0.3	0.95	0.903	0.925 / 0.972	0.3 / 1.0
PushT, Reacher	TwoRoom, Cube	0.3	0.95	0.896	0.901 / 0.935	0.7 / 1.0
PushT, Cube	TwoRoom, Reacher	0.3	0.95	0.912	0.952 / 0.935	0.7 / 1.0
Reacher, Cube	TwoRoom, PushT	0.3	0.95	0.926	0.972 / 0.907	0.7 / 1.0
TwoRoom, PushT, Reacher	Cube	0.3	0.95	0.858	0.802 / 1.000	0.3 / 1.0
TwoRoom, PushT, Cube	Reacher	0.3	0.95	0.889	0.905 / 1.000	0.3 / 1.0
TwoRoom, Reacher, Cube	PushT	0.3	0.95	0.917	0.944 / 0.944	0.3 / 1.0
PushT, Reacher, Cube	TwoRoom	0.3	0.95	0.935	1.000 / 0.869	1.0 / 1.0

G Cross-Stressor Pairs and Boundary Cases

This section reports IR and SR separately for the 24 matched blur/resize checkpoint pairs. Component-wise reporting avoids assigning a continuous joint score to diagnostics with different roles. A pair is labeled behaviorally positive when the augmented checkpoint improves success under the evaluated shift by at least five percentage points while losing no more than five clean-success points relative to its unaugmented counterpart.

Table 10: Descriptive cross-stressor component summary. IR improvement is $1 - \mathrm{IR}_{\mathrm{rel}}$ because the base value is 1; SR change is endpoint SR minus base SR. Spearman values correlate each component change with planning-success change. Binary agreement compares a positive component change with the prespecified positive behavior label. "IR pass / veto" reports absolute endpoint IR passes and the subset rejected by SR. The final cell gives the 2.5th–97.5th percentiles from exact enumeration of all $4^4 = 256$ ordered task-cluster resamples, using linear interpolation. This is a sensitivity distribution, not a new-task population CI; the 24 paired rows are not IID.

Scope	n	ρ_{IR}	ρ_{SR}	binary agreement IR / SR	IR pass / veto	cluster-resampling sensitivity
Pooled	24	0.854	0.913	0.917 / 0.917	6 / 0	IR: [0.263, 0.950] SR: [0.352, 0.929]
Blur	12	0.874	0.953	0.917 / 0.917	3 / 0	–
Resize	12	0.748	0.846	0.917 / 0.917	3 / 0	–
TwoRoom	6	0.543	0.543	1.000 / 1.000	0 / 0	–
PushT	6	0.464	0.603	0.667 / 0.667	0 / 0	–
Reacher	6	0.886	0.886	1.000 / 1.000	6 / 0	–
Cube	6	-0.086	-0.543	1.000 / 1.000	0 / 0	–
LOTO range	18	[0.720, 0.911]	[0.846, 0.901]	–	–	–

The two label disagreements are PushT run 3072 under resize and run 3074 under blur. Each gains four success points, one below the prespecified five-point cutoff, while both diagnostic components move in the favorable direction.

Table 11: Diagnostic ablation on 24 paired blur/resize endpoints. Each metric predicts improvement when its endpoint-to-reference ratio is below one; no threshold is fitted or transferred. Spearman correlation relates diagnostic improvement, defined as one minus this ratio, to planning-success change. Correct/24 and BA compare the predicted sign with the prespecified behavioral label. The LOTO range recomputes pooled Spearman after omitting each task; the 24 pairs are clustered within four tasks.

Diagnostic	Spearman	Correct / 24	BA	LOTO Spearman range
Encoder only	0.859	20	0.778	[0.725, 0.901]
One-step rollout	0.738	20	0.778	[0.454, 0.877]
Zero actions	0.805	21	0.833	[0.626, 0.877]
Cross-anchor action shuffle	0.780	21	0.833	[0.586, 0.869]
Temporal action shuffle	0.844	22	0.889	[0.678, 0.923]
Recorded-action ACPC	0.854	22	0.889	[0.720, 0.911]

H Local Sensitivity under Gaussian Noise

To relate ACPC under Gaussian noise to local model sensitivity, consider a small isotropic perturbation $\delta x \sim \mathcal{N}(0, \sigma^2 I)$. Let J_E and J_G be the Jacobians of the encoder and rollout map, evaluated along the clean input. A first-order expansion gives

$$\Delta G = J_G J_E \, \delta x + o(\|\delta x\|_2).$$

Writing $A = J_G J_E$ and using $\mathbb{E}[\delta x \delta x^\top] = \sigma^2 I$, we obtain

$$\mathbb{E}\big[\|A\delta x\|_2^2\big] = \mathrm{tr}\big(A\,\mathbb{E}[\delta x \delta x^\top] A^\top\big) = \sigma^2\,\mathrm{tr}(AA^\top) = \sigma^2 \|A\|_F^2, \tag{22}$$

Therefore, for small σ,

$$\mathbb{E}[\|\Delta G\|_2^2] \approx \sigma^2 \|J_G J_E\|_F^2.$$

The squared Frobenius norm of the composed Jacobian thus measures how strongly the encoder–rollout path locally amplifies Gaussian observation noise. Jacobian Frobenius norms have previously been used to measure and regularize local representation sensitivity Rifai et al. (2011). We estimate

$$\|J_G J_E\|_F^2 = \mathrm{tr}\big[(J_G J_E)^\top (J_G J_E)\big]$$

with Rademacher probes and the Hutchinson estimator Hutchinson (1989). Jacobian–vector products compute these estimates without constructing the full Jacobian. In every task, both the finite-difference and Jacobian-based estimates are lower for the noise-augmented checkpoint (Figure 8). This result is consistent with Gaussian-noise augmentation reducing ACPC by lowering the local sensitivity of the encoder–rollout path. The analysis applies only near the evaluated inputs and does not provide a global robustness guarantee.

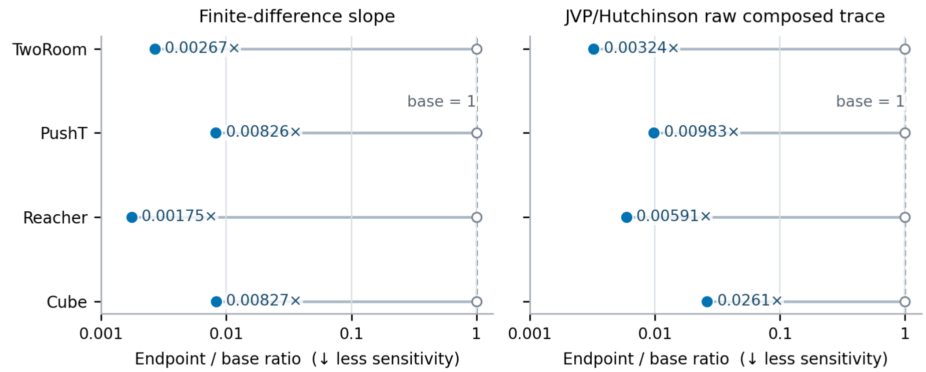

Figure 8: Ratio of local sensitivity for the Gaussian-noise-augmented checkpoint to that of the unaugmented checkpoint. We report a finite-difference estimate and a JVP-based Hutchinson estimate of the composed Jacobian's squared Frobenius norm. Both estimates are below one for every task, indicating lower local sensitivity after augmentation. These local measurements do not establish task performance or global robustness.

I Fixed-Pool Certificates and Adaptive-CEM Selection Regret

The selection-regret analysis does not use task-success labels. We evaluate the unaugmented and $\sigma_{\max} = 0.08$ checkpoints from every task and training run. For each checkpoint, we use 100 histories and apply perturbations at severities 0.02, 0.05, and 0.08. Each clean or perturbed CEM run uses 64 candidate action sequences, eight elite candidates, eight iterations, and a five-step prediction horizon. The paired runs start from the same proposal and use the same random samples, but each run updates its proposal using its own elite candidates.

The implementation averages the squared embedding error across the d embedding coordinates:

$$C_j = \frac{\|x_j - g\|_2^2}{d}.$$

This differs from the summed squared cost in Proposition 2 only by the positive factor $1/d$. The factor rescales every C_j and b_j equally, so the regret bound and selection certificates remain unchanged.

For the fixed-pool certificate analysis, we capture CEM's ordered first-round proposal of 64 candidates for each history and evaluate that same pool from the clean and perturbed inputs. At each severity, we average the rates equally over 12 task–training-run blocks, with 100 histories per block. The same histories recur across severities and checkpoint roles, so the entries in Table 12 are paired descriptive rates rather than independent observations. The candidate-level cost bound is evaluated directly from the two predicted endpoints and the fixed goal, as in Equation (4).

Table 12: Empirical coverage of the candidate-specific planning-cost certificates on the fixed first-round CEM pool. Each entry is the equal-weight mean over 12 task–training-run blocks, with 100 histories and the same 64 candidates per clean–perturbed pair. "Top-1 cert." and "Elite cert." are the fractions satisfying Equation (12) and Equation (13), respectively; "Same top-1" is the observed fraction whose lowest-cost candidate is unchanged. The zero false certificates and zero cost-bound violations are deterministic implementation checks. These results concern one fixed candidate pool, not subsequent adaptive CEM iterations or simulator return.

Checkpoint	Probe σ	Top-1 cert. (%)	Elite cert. (%)	Same top-1 (%)
Unaugmented	0.02	2.3	0.0	67.4
Unaugmented	0.05	0.0	0.0	20.3
Unaugmented	0.08	0.0	0.0	8.7
Augmented ($\sigma_{\max} = 0.08$)	0.02	69.5	30.4	98.2
Augmented ($\sigma_{\max} = 0.08$)	0.05	46.5	14.0	96.3
Augmented ($\sigma_{\max} = 0.08$)	0.08	34.8	8.8	93.3

For each CEM run, we compute ACPC for every candidate under that candidate's action sequence. We summarize the candidate pool by the q90 ACPC at horizons one and five. Let $\mathbf{a}$ be the plan selected from the clean history and $\tilde{\mathbf{a}}$ the plan selected from its perturbed counterpart. The regression target is the extra predicted cost of the perturbed-history plan when both plans are evaluated from the clean history:

$$\left[C_h(\tilde{\mathbf{a}}) - C_h(\mathbf{a})\right]_+.$$

Within each training run, we fit ridge regressions on three tasks and evaluate them on the remaining task, rotating through all four choices. MAE is computed on $\log(1 + \text{selection regret})$ and averaged equally across the four evaluation tasks. The relative MAE reduction is computed separately for each training run before reporting the mean and sample standard deviation across the three runs. This experiment evaluates whether ACPC reflects the model-cost effect of a CEM selection change; it does not evaluate simulator return.